Lecture 20

Fibre bundles

Objectives. *A vector field assigns a vector in a different tangent space at each point. A bundle describes these fibres and the changes between their local coordinates. We reconstruct bundles from transition functions and prove the global-section criterion for principal bundles. The Hopf bundle and the tangent sphere show how local products can fail to give a global product.*

By the end of this lecture you will be able to:

- Construct a smooth vector or principal bundle from a smooth cocycle.
- Relate local frames, section components and transition functions.
- Prove the principal-bundle section criterion and identify the Hopf bundle.
- Distinguish the tangent sphere's frame winding from its component winding.

20.1 Local products and fibres

A smooth *fibre bundle* consists of a smooth surjection $\pi_E : E \to M$ and a manifold F, its *typical fibre*. Every point of the *base space* M has an open neighbourhood U_i and a diffeomorphism

$$\varphi_i : U_i \times F \longrightarrow \pi_E^{-1}(U_i), \qquad \pi_E(\varphi_i(x, v)) = x.$$

This is a *local trivialization*. The manifold E is the *total space*. The *fibre at* x is $E_x = \pi_E^{-1}(x)$. All manifolds here are Hausdorff and second countable. A *bundle isomorphism* over M is a diffeomorphism of total spaces commuting with the projections and preserving the specified fibre structure. A bundle is *trivial* if it is isomorphic to $M \times F$. The figure shows a local product only.

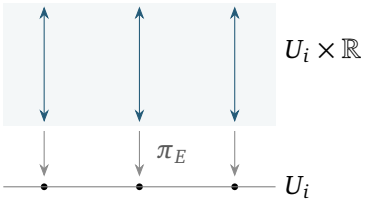


Figure 20.1: A local trivialization gives fibre coordinates over U_i. The projection sends each vertical fibre to its base point.

A real or complex *vector bundle of rank* r has vector-space fibres and local trivializations linear on each fibre, with typical fibre $V_{\text{rep}} = \mathbb{R}^r$ or $\mathbb{C}^r$. A rank-one vector bundle is a *line bundle*. For example, TM of Lecture 8 is a rank-dim M real vector bundle.

A *section over an open set* $U \subset M$ is a smooth map $s : U \to E$ such that $\pi_E \circ s = \mathrm{id}_U$. It is a *global section* when $U = M$. A *local frame* of a rank-r vector bundle is a row of r smooth sections that form a basis in each fibre of its domain.

Let G be a Lie group. A *principal G-bundle* is a bundle $\pi_E : P \to M$ with a smooth right action of G that is free and transitive on each fibre. Its local trivializations are required to satisfy $\varphi_i(x, ab) = \varphi_i(x, a)b$.

20.2 Transition functions and reconstruction

For a vector bundle, comparing trivializations defines smooth *transition functions* $g_{ij} : U_i \cap U_j \to \mathrm{GL}(r, \mathbb{K})$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, by

$$\varphi_j(x, v) = \varphi_i(x, g_{ij}(x)v).$$

Thus the coordinates of the same vector satisfy $v_i = g_{ij}v_j$. The *structure group* is the group of allowed transition matrices, usually $\mathrm{GL}(r, \mathbb{R})$ or $\mathrm{GL}(r, \mathbb{C})$. For a principal bundle, equivariance instead gives

$$\varphi_j(x, a) = \varphi_i(x, g_{ij}(x)a), \qquad g_{ij} : U_i \cap U_j \to G.$$

Indeed, compare the trivializations at the group identity and then act on the right by a. This also shows that the principal transitions are smooth. For a vector bundle, the standard bases give local row frames e_i with $e_j = e_i g_{ij}$. On triple overlaps the *cocycle condition* is

$$g_{ij}g_{jk} = g_{ik}, \qquad g_{ii} = I, \qquad g_{ji} = g_{ij}^{-1}.$$

For a principal bundle, replace the identity matrix I by the group identity e. The figure follows the coordinate maps, whose directions reverse the frame relation.

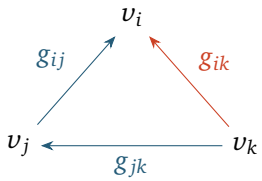


Figure 20.2: A fibre-coordinate column passes from region k to j and then to i by $g_{ij}g_{jk}$. Direct passage uses g_{ik}. All maps refer to the same base point in a triple overlap.

Proposition 20.1 (Cocycle reconstruction · *proved*)**.** *Smooth transition matrices satisfying the cocycle condition on an open cover of M determine a smooth vector bundle, up to bundle isomorphism. The same construction with fibre a Lie group G and transition functions acting by left multiplication constructs a principal bundle. Let $h_i : U_i \to \mathrm{GL}(r, \mathbb{K})$ be smooth frame changes. For a principal bundle take smooth $h_i : U_i \to G$ instead. In this case replace frames by local sections in the following change law. Under changes $e'_i = e_i h_i$, the transitions become*

$$g'_{ij} = h_i^{-1} g_{ij} h_j.$$

Conversely, transitions related in this way give isomorphic bundles. Different covers can be compared on their common refinement.

Proof. Take the disjoint union of $U_i \times V_{\mathrm{rep}}$ and identify

$$(x, v)_j \sim (x, g_{ij}(x)v)_i.$$

Identity, inverse and product in the cocycle make this an equivalence relation. Project its quotient to M by the common base point. Over U_i, each class has a unique representative in

the ith copy. The corresponding local coordinate map sends a representative in copy j to $(x, g_{ij}(x)v)$. It is a homeomorphism because the preimage of an open set in one copy is open in every other copy, since the transition map is a diffeomorphism. The quotient topology therefore agrees with these product charts. Their changes are smooth and linear on fibres.

Points with different base points have disjoint neighbourhoods lifted from M. Points with the same base point can be separated in one product chart. Thus the total space is Hausdorff. Second countability of M gives a countable subcover. Choose a basis element inside a cover member at each point and choose one member per basis element. The corresponding product charts, each second countable, give a countable basis for the total space. Hence the quotient is a smooth manifold.

The same argument with V_{rep} replaced by G gives smooth charts $U_i \times G$. The action $(x, a) \cdot b = (x, ab)$ is independent of the chart because left transition multiplication commutes with right multiplication. It is free and transitive on each fibre, as required in the principal definition.

Finally, $e'_j = e'_i h_i^{-1} g_{ij} h_j$, proving the displayed change law. For an isomorphism between the two constructions, use $(x, v)_i \mapsto (x, h_i^{-1} v)'_i$. The change law makes this map agree on overlaps. Its inverse uses h_i, so it is a smooth bundle isomorphism. If another bundle has trivializations φ_i with these transitions, send the quotient class $[(x, v)_i]$ to $\varphi_i(x, v)$. The defining transition relation makes this independent of its representative. In product charts it is the identity, so it is a smooth bundle isomorphism, equivariant in the principal case. This proves uniqueness. Restricting trivializations from two covers to their pairwise intersections gives a common cover. The same chartwise map compares the reconstructions on this refinement. ∎

This is the smooth reconstruction of Nakahara, §9.2.2, with our frame convention. Naber's Theorem 4.3.4, pp. 223–224, is its continuous principal-bundle counterpart, with the same transition convention.

20.3 Sections and frames of a vector bundle

In a local frame, write $s = e_i \sigma_i$. Its coefficient columns satisfy

$$\sigma_i = g_{ij} \sigma_j.$$

Conversely, smooth columns with this compatibility glue to a section. The zero columns always give a global zero section.

A global frame exists exactly when the vector bundle is trivial. Indeed, a frame defines the isomorphism $(x, v) \mapsto e(x)v$. Its inverse is smooth in local coordinates because the frame matrix is invertible. A trivialization supplies the converse frame by its standard basis. For a line bundle, one nowhere-zero section is a frame. For higher rank, one section need not give an entire basis.

20.4 Principal and associated bundles

The local sections $s_i(x) = \varphi_i(x, e)$ obey $s_j = s_i g_{ij}$. The group acts on the right, while g_{ij} changes fibre coordinates by multiplication on the left.

> **Proposition 20.2** (The section criterion · *proved*)**.** *A principal bundle is trivial if and only if it has a global section.*

Proof. A global section s gives a map $\Psi\colon M\times G\to P$ by $\Psi(x,a)=s(x)a$. Freeness and transitivity make it bijective and equivariant. In a local trivialization write $s(x)=s_i(x)b_i(x)$. Then $\Psi(x,a)$ has local coordinate $b_i(x)a$. Its inverse has coordinate $b_i(x)^{-1}a$, so both maps are smooth. Conversely, a trivialization gives the section $x\mapsto(x,e)$. ∎

This is Nakahara, Theorem 9.2. Vector-bundle triviality requires a full frame, as above.

For an oriented Riemannian n-manifold, its *oriented orthonormal frame bundle* consists of all positive orthonormal frames in its tangent spaces. Right multiplication of a row frame by $\mathrm{SO}(n)$ is free and transitive on each fibre. Local Gram–Schmidt frames give equivariant product charts. This makes it a principal $\mathrm{SO}(n)$-bundle, even when no global tangent frame exists.

Remark 20.3 (Associated bundles)**.** Let $\rho\colon G\to\mathrm{GL}(V_{\mathrm{rep}})$ be a smooth representation. The *associated bundle* is

$$P\times_G V_{\mathrm{rep}}=(P\times V_{\mathrm{rep}})/{\sim},\qquad (pa,v)\sim(p,\rho(a)v),\qquad [p,v]\longmapsto\pi_E(p).$$

Its local trivializations are $(x,v)\mapsto[s_i(x),v]$. Their transitions are $\rho(g_{ij})$, so reconstruction gives a smooth vector bundle. Sections correspond to smooth *equivariant maps* $f\colon P\to V_{\mathrm{rep}}$ with

$$f(pa)=\rho(a)^{-1}f(p).$$

The section is $x\mapsto[p,f(p)]$ for any p over x: $[pa,f(pa)]=[p,\rho(a)f(pa)]=[p,f(p)]$. Conversely, a section uniquely determines these fibre coordinates $f(p)$. In a product chart they are $\rho(a)^{-1}f(s_i(x))$, which proves smoothness and the converse. The homogeneous case is the construction in *Lie Groups*, Lecture 26, *From equivariant functions to sections*.

20.5 The Hopf bundle

For a closed subgroup H of a Lie group G, Theorem 8.5 gives the smooth quotient G/H and a submersion $\pi_E\colon G\to G/H$. The inverse function theorem gives local submersion coordinates (Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.5). Choose base coordinates y and supplement the independent differentials of $y\circ\pi_E$ by complementary coordinates ζ. The map $(y\circ\pi_E,\zeta)$ has invertible differential, hence is a local coordinate system. In these coordinates $\pi_E(y,\zeta)=y$. Fixing ζ gives a smooth local section s near each base point. The maps

$$U\times H\longrightarrow\pi_E^{-1}(U),\qquad (x,h)\longmapsto s(x)h$$

are diffeomorphisms. Indeed, their inverse sends a to $(aH,s(aH)^{-1}a)$. The second entry lies in the embedded subgroup H and is smooth there. In submanifold charts its normal coordinates vanish. Thus $H\to G\to G/H$ is a principal H-bundle.

Let $N=[1:0]$ and $S=[0:1]$ be the north and south poles of the Bloch sphere. The projective coordinate $z=b/a$ is defined on $U_N=\mathbb{C}\mathrm{P}^1\setminus\{S\}$. The coordinate $w=a/b=1/z$ is defined on $U_S=\mathbb{C}\mathrm{P}^1\setminus\{N\}$. Unit representatives of their lines are

$$p_N(z)=\frac{(1,z)^T}{\sqrt{1+|z|^2}},\qquad p_S(w)=\frac{(w,1)^T}{\sqrt{1+|w|^2}}.$$

The *tautological line bundle* is the set

$$\mathcal{T}=\{(\ell,v)\in\mathbb{C}\mathrm{P}^1\times\mathbb{C}^2:v\in\ell\},\qquad (\ell,v)\longmapsto\ell.$$

Its local coordinates are $v = cp_N(z)$ or $v = cp_S(w)$, with $c \in \mathbb{C}$. The inverse coefficient is $c = p_N^\dagger v$ or $c = p_S^\dagger v$, respectively. These maps give smooth product charts with linear overlap maps. On the overlap, $p_S = p_N|z|/z$. Orient the equator $|z| = 1$ by increasing ϕ, as the boundary of the northern cap. On this positive equator $z = e^{i\phi}$, the frame transition is $g_{NS} = e^{-i\phi}$, with winding -1. This sign follows from the stated frame order.

Theorem 20.4 (The Hopf bundle · *proved given the cited homogeneous-space and inverse function theorems*)**.** *The map* $\mathrm{SU}(2) \to \mathrm{SU}(2)/\mathrm{U}(1) \cong \mathbb{C}\mathrm{P}^1 \cong S^2$ *is a nontrivial principal* $\mathrm{U}(1)$*-bundle. Under* $\mathrm{SU}(2) \cong S^3$*, it sends a unit column* $(a, b)^T$ *to its line* $[a : b]$*. Its associated complex line bundle for the standard* $\mathrm{U}(1)$ *representation is the tautological line bundle.*

Proof. Recall Lecture 8's explicit identification

$$(a, b) \longmapsto \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Right multiplication by $\mathrm{diag}(u, u^{-1})$, for $u \in \mathrm{U}(1)$, multiplies the first column by u. Its orbits are exactly the unit representatives of each complex line. The preceding quotient argument proves the principal structure.

To prove that S^n is simply connected for $n \in \{2, 3\}$, subdivide a loop into hemispherical pieces using Lemma 3.5. On each piece replace it by the normalized straight segment between its endpoints. Both curves have positive scalar product with the hemisphere centre, so their normalized straight interpolation is a homotopy fixing those endpoints. The resulting polygon lies in finitely many two-dimensional linear subspaces. Their union misses a point of the sphere. For each subspace choose a nonzero linear functional vanishing on it and evaluate these on $(1, t, \dots, t^n)$. Each gives a nonzero polynomial, so a parameter outside their finite root sets gives a vector outside the union. Normalize it to obtain the omitted sphere point. Stereographic projection from that point puts the polygon in $\mathbb{R}^n$, where straight contraction to its base point fixes that base point. Pasting the piece homotopies proves the assertion for $n = 2$ and $n = 3$. Both spheres are path-connected by short arcs, joining antipodes via a third point.

Loops and based homotopies in a product are precisely pairs of loops and homotopies. Thus $\pi_1(S^2 \times S^1) \cong \mathbb{Z}$, whereas $\pi_1(S^3) = 0$. A trivial Hopf bundle would give a diffeomorphism between these total spaces, which is impossible.

The map $[p, v] \mapsto ([p], vp)$ identifies the associated line with $\mathcal{T}$. It is well-defined because $(pu, v) \sim (p, uv)$. Its inverse exists on each fibre, and in the local unit representatives above both maps are product-coordinate identity maps. Hence it is a smooth bundle isomorphism. ∎

The figure separates the phase fibre from its fixed base line.

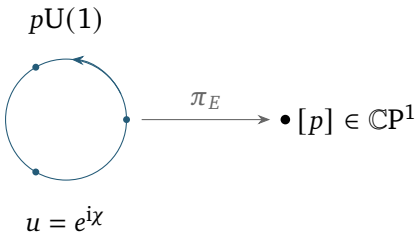


Figure 20.3: All unit vectors pu, with $u \in \mathrm{U}(1)$, lie over the same line $[p]$. The circle records their relative phase.

Proposition 20.5 (Clutching classification · *cited*). *The underlying topological principal* $\mathrm{U}(1)$*-bundles over* S^2*, up to continuous equivariant bundle isomorphism, are classified by the winding of the transition* g_{NS} *on the positive equator. Every integer occurs.*

This is Naber, Theorem 4.4.3, pp. 231–232. That theorem concerns continuous bundles. No smooth classification proof is claimed here. Its integer invariant already distinguishes the smooth examples. On the overlap $z \neq 0$, the transitions $(z/|z|)^n$ give smooth representatives for every $n \in \mathbb{Z}$.

20.6 The tangent sphere and its transition winding

Use the oriented complex charts $z = x + \mathrm{i}y$ and $w = u + \mathrm{i}v = 1/z$ of Lecture 7 on $S^2 \cong \mathbb{C}\mathrm{P}^1$. Multiplication by i on tangent vectors is defined by $\mathrm{i}\partial_x = \partial_y$. The change $w = 1/z$ has complex-linear differential and preserves this rule. The real tangent planes therefore form a complex line bundle.

The figure displays the two angle lifts computed in the example.

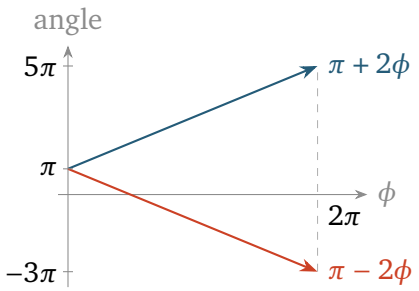


Figure 20.4: The frame angle increases by 4π during one equatorial circuit. The inverse component angle decreases by 4π.

Example 20.6 (The tangent-sphere transition). Find its frame transition and its component transition around the equator. Use complex frames $e_z = \partial_x$ and $e_w = \partial_u$, where a complex coefficient $a + \mathrm{i}b$ means the real vector $a\partial_x + b\partial_y$ in the first frame. The coordinate derivative and its inverse are

$$\frac{\mathrm{d}w}{\mathrm{d}z} = -z^{-2}, \qquad \frac{\mathrm{d}z}{\mathrm{d}w} = -z^2.$$

The chain rule gives the frame relation and the component relation

$$e_w = e_z(-z^2), \qquad \sigma_w = (-z^{-2})\sigma_z.$$

The real plane rotation through β is

$$\mathrm{Rot}(\beta) = \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix}.$$

Identify a complex section coefficient with the real column $(\mathrm{Re}\,\sigma, \mathrm{Im}\,\sigma)^T$. On $z = e^{\mathrm{i}\phi}$ the corresponding real matrices are

$$g_{zw} = -\mathrm{Rot}(2\phi), \qquad \sigma_w = -\mathrm{Rot}(-2\phi)\sigma_z.$$

The frame loop has lifted angle $\pi + 2\phi$ and winding +2. The component loop has angle $\pi - 2\phi$ and winding −2. The Bloch map is $(2x, 2y, 1 - |z|^2)/(1 + |z|^2)$. For the round metric,

differentiating it gives

$$g = \frac{4(\mathrm{d}x^2 + \mathrm{d}y^2)}{(1 + |z|^2)^2}.$$

Normalizing the coordinate frames therefore replaces $-z^2$ by its phase $-z^2/|z|^2$ on the overlap. Its equatorial winding is still +2. A new unit frame on either cap changes this winding by zero, because its boundary phase extends across the disk and is null-homotopic by Lecture 3. Consequently the oriented orthonormal frame bundle is nontrivial by the cited classification. A nowhere-zero tangent field would give a unit frame and trivialize it, contradicting its winding. The integer 2 agrees with $\chi(S^2)$ in Lecture 18.

Self-check. Describe the Möbius real line bundle over S^1 and show it is nontrivial.

Answer. Take $[0, 2\pi] \times \mathbb{R}$ and identify $(2\pi, v)$ with $(0, -v)$. On two overlapping base arcs the transition is +1 on one overlap component and −1 on the other. A nowhere-zero section would lift to a continuous real function with $f(2\pi) = -f(0)$, impossible by the intermediate value theorem. A trivial line bundle has a nowhere-zero section, so this bundle is nontrivial. Its zero section still exists. Putting −1 on both overlap components instead gives a trivial bundle. Change the frame on one whole arc by −1 to make both transitions +1.

Summary. Transition functions reconstruct smooth vector and principal bundles. Frames transform by $e_j = e_i g_{ij}$, while section components transform inversely. A principal bundle is trivial exactly when it has a global section. The Hopf bundle $S^3 \to S^2$ is nontrivial and its associated standard line is tautological. The tangent sphere has frame winding +2, opposite to its component winding. Lecture 21 describes connections and transport in these local frames.

Reading. Nakahara, §9.2–9.4, especially Theorem 9.2. Naber, §§4.2–4.4, especially Theorems 4.3.4 and 4.4.3.